HSLU · DATA-DRIVEN SUPPLY CHAIN MANAGEMENT

Inventory Optimisation – Final Report

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Code and notebooks: github.com/divyashori-datascience/scm-inventory-optimisation

Presentation: presentation/Inventory_Optimisation_Final_Presentation.pdf

Executive summary

The company can release about 1.27M of tied-up capital (−26% vs 2021) while raising availability to 95% fill, by stocking only the right articles and setting their reorder points from demand every month.

Capital lockup rose from 16% of revenue (2019) to 27% (2022). We analysed 2.69M stock movements (2007–2022, 58,764 articles) in three steps that build on each other:

Step	Question	Answer
1 · Data analysis	Where does the capital sit and how healthy is it?	Stock in inventory articles +39% since 2018; 37% of stock value has no reorder point; 0.94M idle for over a year
2 · Classification	Which articles belong on stock?	Four classes; Runners (10% of active articles, 74% of usage) go on stock, Repeaters selectively, One-offs and Dormant articles do not
3 · Reorder points	At what level should each stock article reorder?	Demand-driven levels, recalculated monthly: 95% fill (97% on A-articles) with 26% less capital than 2021

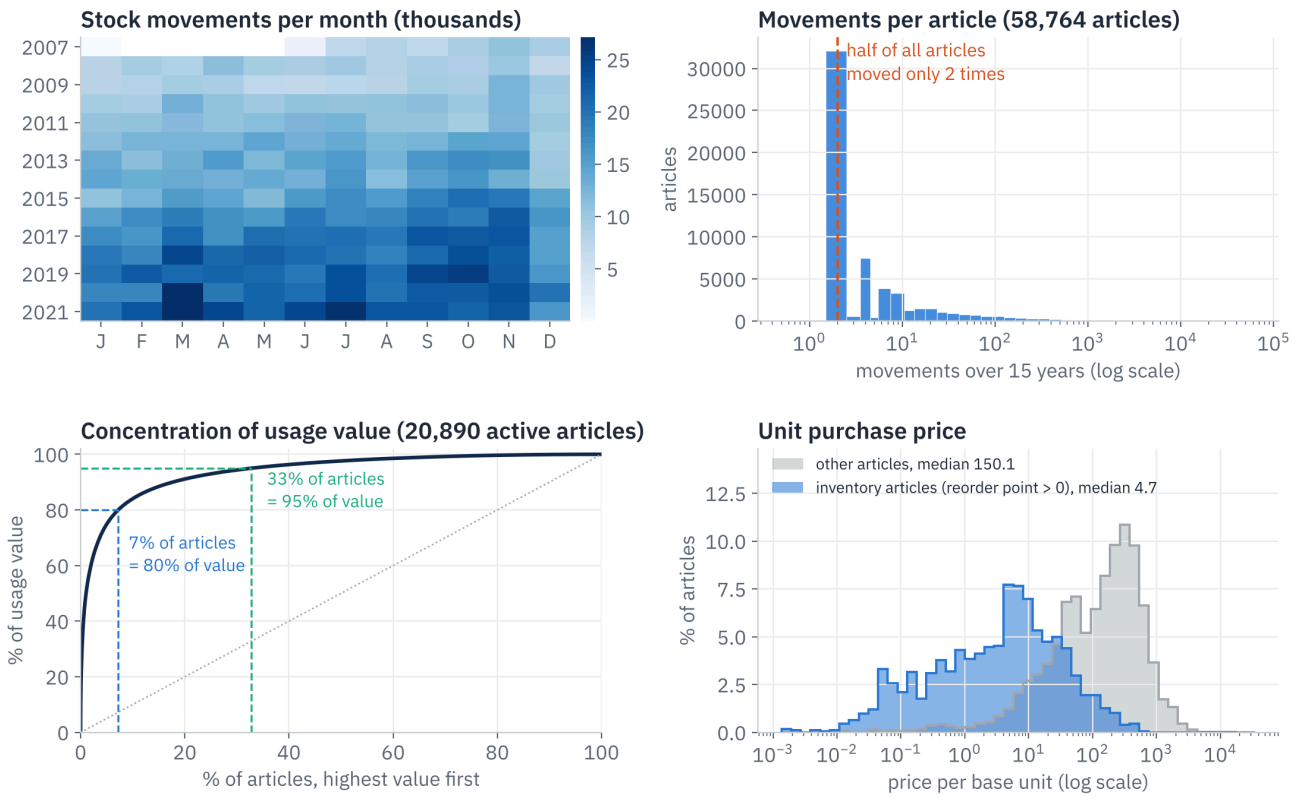
Recommended strategy: clean the range (de-list 427 stock articles, run down idle stock), stock by class, and replace manual reorder points with demand-driven ones recalculated every month, starting with the A-articles that hold about 65% of capital. All values are in the dataset's anonymised currency units at current purchase price.

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1 · Data and cleaning approach

Every number in this report comes from a daily stock history we rebuilt per article from 2.69M movements, after a data audit found four issues that would otherwise distort capital and demand.

The data at a glance



Top left: stock movements per month 2007–2021. Top right: movements per article, log scale. Bottom left: concentration of usage value (Lorenz curve). Bottom right: unit purchase price, inventory articles vs others

- **Activity grew about five-fold:** 108k movements in 2008, 274k in 2021. Rules set years ago are unlikely to fit today.

- **A very long tail:** half of all 58,764 articles moved only twice in fifteen years (typically one receipt, one use).
- **Value is concentrated:** 7% of the active articles carry 80% of usage value, 33% carry 95%. This is the basis for the ABC service levels in section 6.
- **Stock articles are cheap, high-volume parts:** median unit price 4.7 for articles with a reorder point, 150 for the rest.

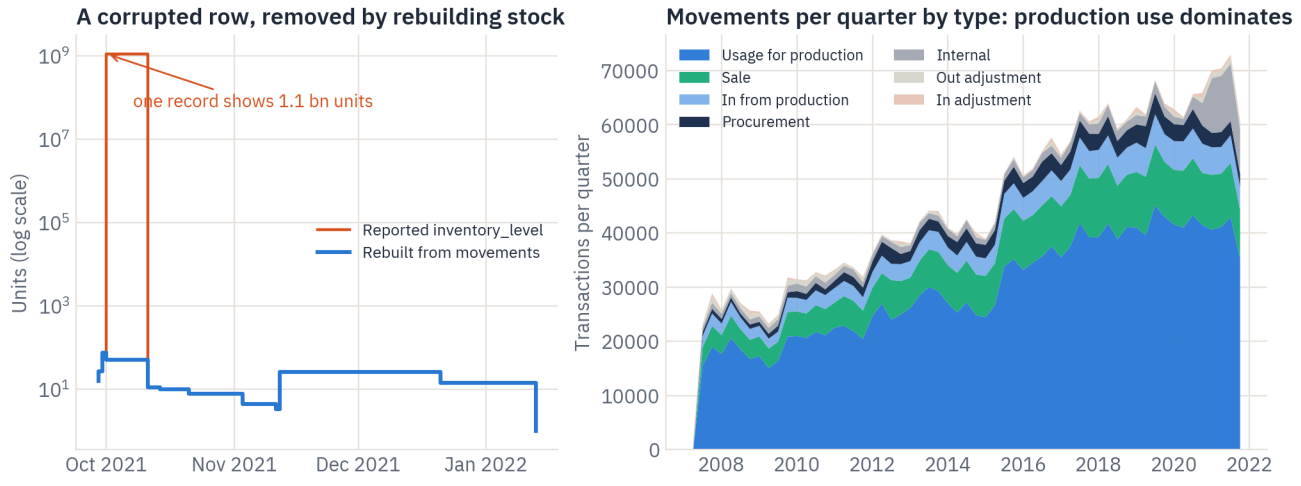
Preparation pipeline

Step	Count	What happens
Raw movements	2,694,849	17 attributes, 2007–2022, no missing values
Stock rebuilt	160 levels repaired	stock = opening level + cumulative movements
Demand movements	2.18M	sales (1) + production use (5), returns netted
Daily table	58,764 articles × 5,480 days	stock and demand per article and day
Active articles	20,890	demand in the last three years → Part 2
Inventory articles	1,751	reorder point > 0 → Part 3

Dataset: 2,694,849 transactions, 17 attributes, 58,764 articles, 21.01.2007 – 21.01.2022, no missing values. 1,751 articles carry a safety stock above zero; the company uses that field as the **reorder point**.

Audit finding	Evidence	What we did
`inventory_level` is the stock **before** each movement, not after as documented	99.99% of rows satisfy next level = level + quantity	Rebuilt stock as opening level + cumulative movements
Corrupted rows	160 rows inconsistent with the movement history; one shows 1.1 bn units	Removed automatically by the rebuild
Quantities are anonymised	100% of quantities are integer multiples of 1.1107; safety stock and MOQ scaled the same way	Kept as delivered; ratios between quantity fields stay valid
Master data is a 2022 snapshot	Safety stock, lead time, MOQ, price never change within an article	History valued at today's price, compared with today's settings

Demand definition: net sales (type 1) plus usage for production (type 5). Returns are netted out (3% of sales rows are positive, 28% of procurement rows negative). Internal transfers (4) and adjustments (2, 3) are not demand.

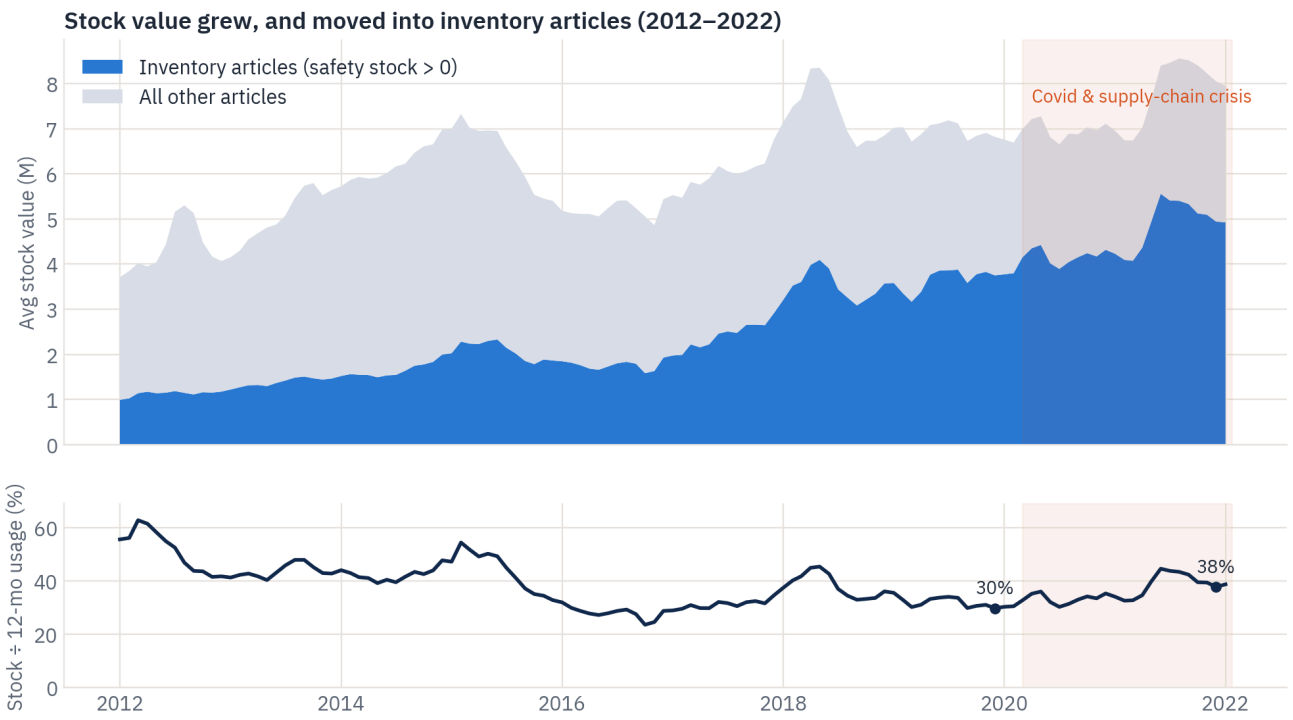


Left: one article's reported vs rebuilt stock on a log scale. Right: quarterly transactions by movement type, 2007–2021

The left panel shows why rebuilding matters: one corrupted record would put a single cheap article at 1.1 bn units and inflate 2021 capital from about 8M to almost 2.9 bn. The right panel shows what the business is: 65% of all movements are usage for production, so most stock feeds the company's own production, not retail sales.

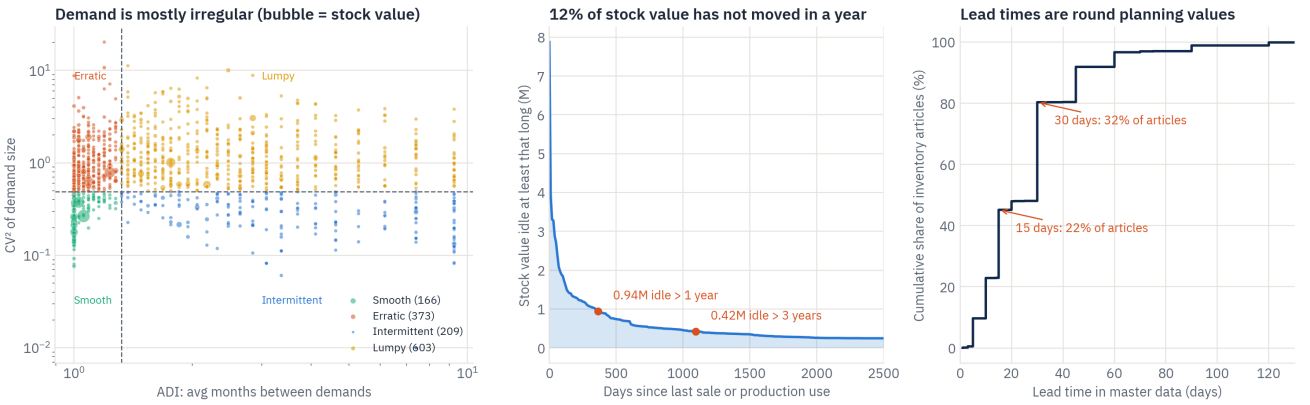
2 · What the data tells us

Capital is growing faster than demand, a large part of it is not managed by any rule, and the rules that exist do not fit the demand they face.



Top: monthly average stock value 2012–2022, inventory articles vs all others. Bottom: stock value of inventory articles divided by their 12-month usage value

- **Capital moved into inventory articles.** Their stock value rose from 3.5M (2018) to 4.9M (2021), **+39%**, and their share of all stock from 48% to 62%. Stock grew faster than usage: from 30% to 38% of a year's usage.
- **37% of stock value (2.9M of 7.9M) sits in articles without a reorder point.** That is leftover or speculative stock no replenishment rule controls, a key input for the classification.



Left: demand patterns of inventory articles (Syntetos–Boylan). Middle: stock value idle for at least x days. Right: cumulative distribution of lead times

- **Demand is mostly irregular.** Only 166 inventory articles (about 10%) have smooth demand; the rest are erratic, lumpy, intermittent or sporadic, the case's *searched demand*. On 5% of demand days, rare large orders carry 29% of volume. One textbook safety-stock formula cannot fit all articles.
- **12% of stock value (0.94M) has not moved for a year**, 0.42M of it for three years or more. 318 inventory articles keep a reorder point despite no demand for over a year.
- **Lead times are manual planning values.** They are maintained for 98% of inventory articles, but 54% sit on exactly 15 or 30 days. There are no order dates, so the real lead time and its variability cannot be measured.

What this means for the study: the classification must decide which articles deserve stock at all (Part 2), and reorder points must follow each article's own demand pattern, separate rare large orders and be tested for lead-time errors (Part 3).

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3 · Classification approach

Three classifications, three decisions. One grouping cannot answer three different questions, so the project uses three, each for one decision:

Classification	Question	Based on	Groups	Used for
Clusters (Part 2)	Should the article be on stock at all?	demand behaviour, K-means	Runners, Repeaters, One-offs, Dormant	stock, stock selectively, buy to order, phase out
Demand pattern (Part 3)	Which reorder-point formula fits?	ADI and CV^2 (Syntetos–Boylan)	Smooth, Erratic, Intermittent, Lumpy, Sporadic	normal formula vs empirical lead-time quantile
ABC (Part 3)	How much safety buffer?	12-month usage value, cut at 80% / 95%	A, B, C	fill target 98% / 95% / 90%

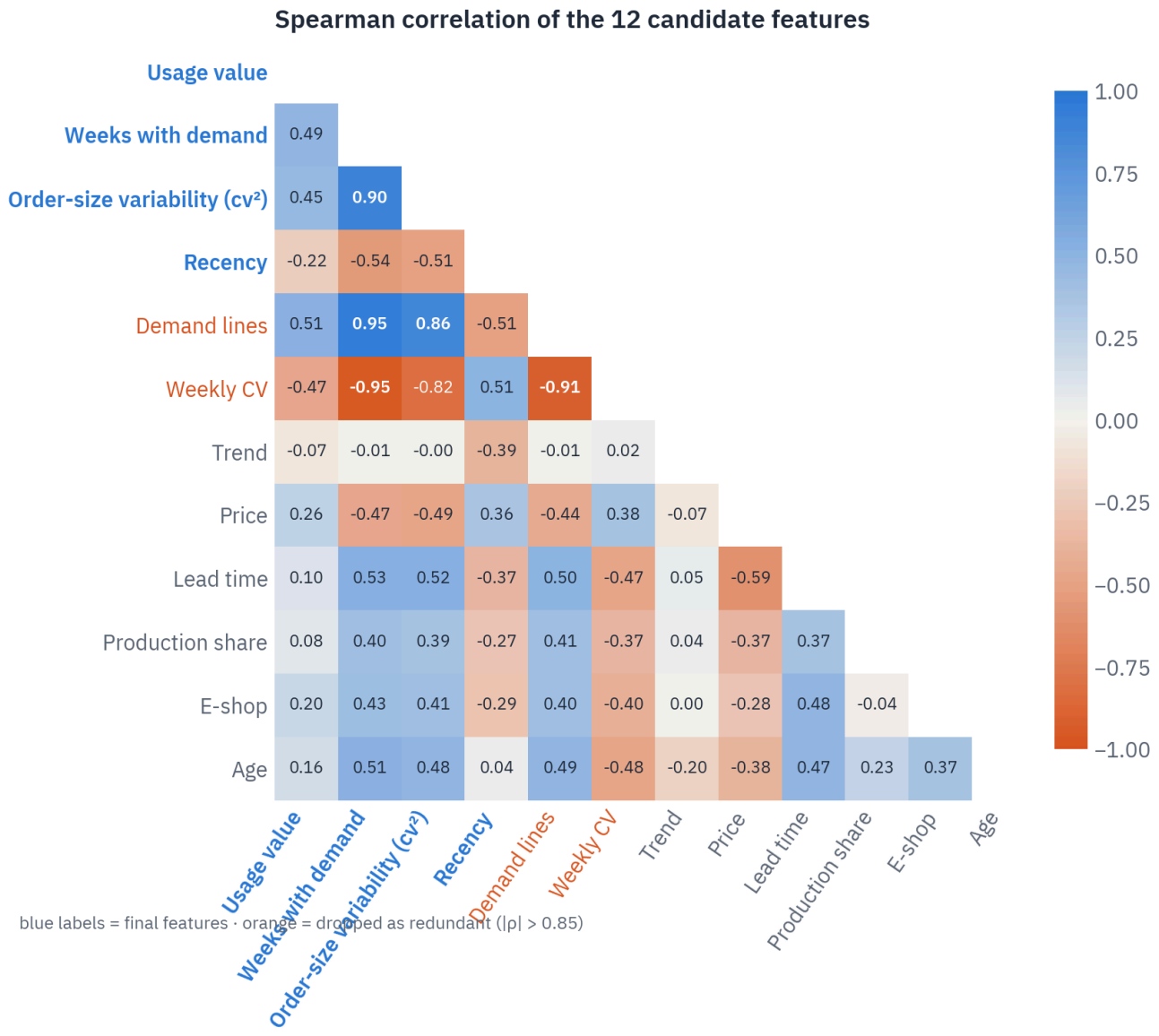
In one line: the cluster decides *if* we stock an article, the demand pattern *how* we compute its reorder point, ABC *how safely*.

We classified all 20,890 articles with demand in the last three years using K-means on four demand features; it beat two alternative algorithms on every quality metric and is highly stable.

Population. Every article with at least one sale or production use between 22.01.2019 and 21.01.2022, stocked or not, because the question is *which* articles should be on stock. Of these, 1,499 are stocked today.

Feature selection (12 → 10 → 4).

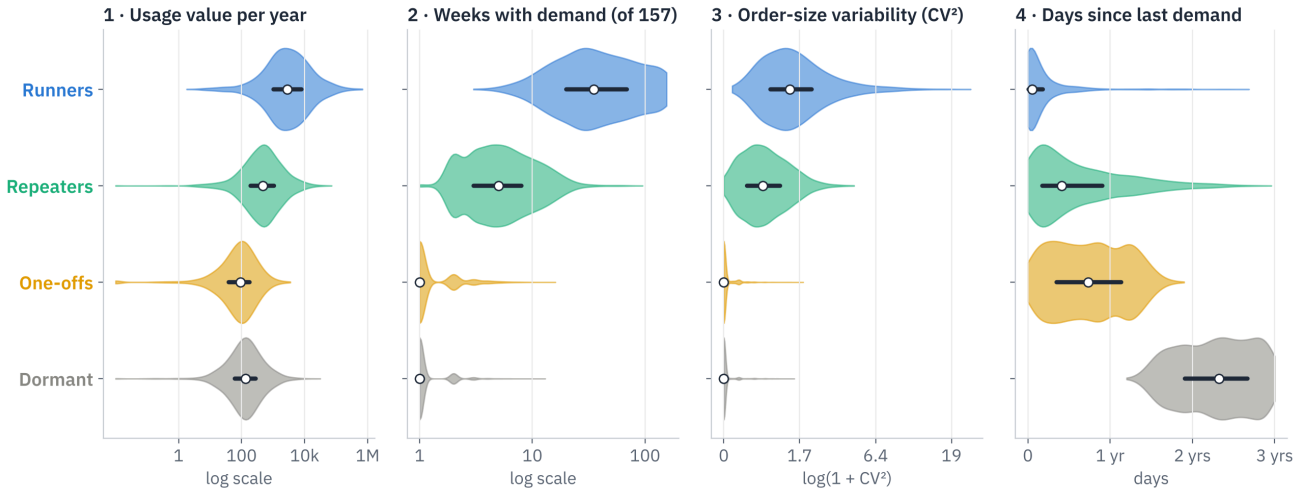
1. Twelve candidate features describe value, frequency, variability, recency, price, lead time, trend, product type, e-shop and age.
2. A redundancy filter drops one of every pair with Spearman $|\rho| > 0.85$: demand lines and weekly CV go. Order-size variability (cv^2) is kept: its $\rho = 0.90$ with frequency is an artefact of one-time demand ($cv^2 = 0$ by definition) and falls to 0.49 for articles with three or more demand weeks.
3. A domain screen keeps four features: price is already inside usage value, lead time leaks today's stocking decision, product type, e-shop and age only profile the classes, and trend is noisy for low-volume articles.



Lower-triangle Spearman correlation matrix of the 12 candidate features; final features in blue, redundant features in orange

Final features: annual usage value (log), weeks with demand (log), order-size variability cv^2 (log), days since last demand; all standardised.

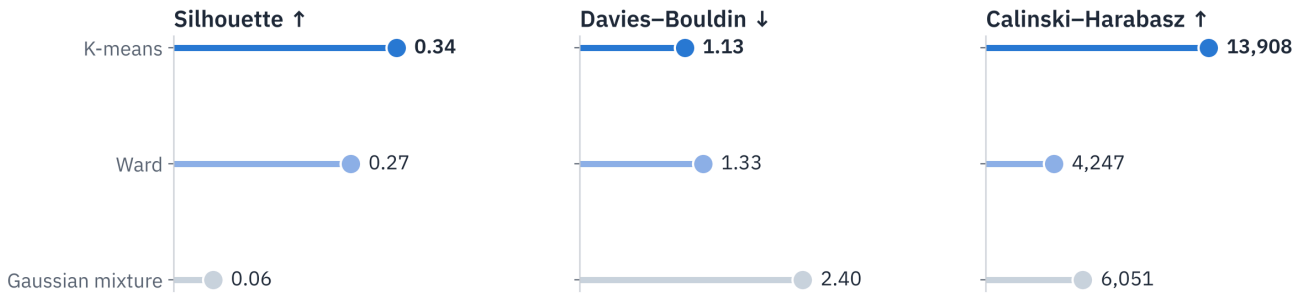
Feature	Definition	Why it matters for stocking
Usage value	units used per year × purchase price	how much money flows through the article
Weeks with demand	weeks with any demand, of 157	how regularly it is needed
Order-size variability	CV^2 of non-zero order sizes	how predictable each order is
Recency	days since last demand	whether the article is still alive



Violin plots of the four features per class; dot = median, bar = middle 50% of articles

Each class has a clear fingerprint: weeks with demand separates Runners and Repeaters from the rest, recency separates One-offs from Dormant articles.

Algorithm, number of classes and validation.



Three quality metrics for K-means, Ward and a Gaussian mixture with four classes

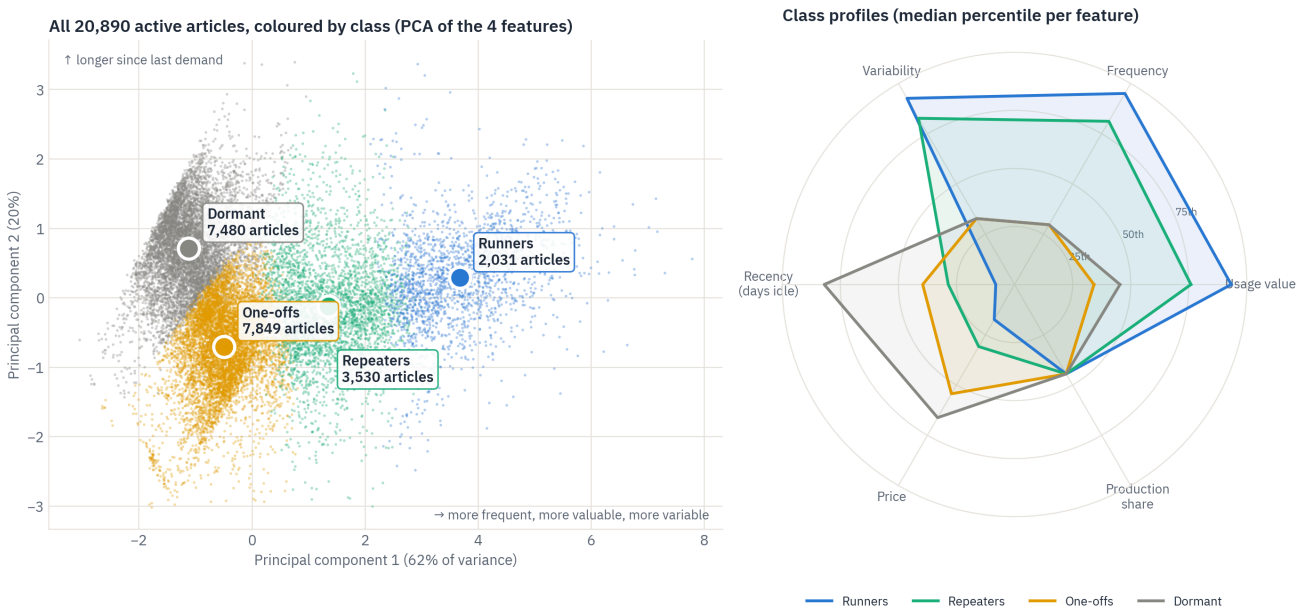
Check	Result
K-means, k = 4	Silhouette 0.34 · Davies–Bouldin 1.13 · Calinski–Harabasz 13,908
Agglomerative (Ward), k = 4	0.27 · 1.33 · 4,247 (8,000-article sample)
Gaussian mixture, k = 4	0.06 · 2.40 · 6,051
Choice of k	k = 3 scores marginally higher (0.35) but merges frequent and occasional articles that need different strategies
Stability	Bootstrap adjusted Rand index 0.98 (min 0.96) over 20 refits
Benchmark	Classic XYZ puts 98% of articles in "Z", so ABC-XYZ cannot separate this range
What separates the classes	Random forest, permutation importance: recency > variability > usage value > frequency

Pure silhouette maximisation was tested and rejected: it kept the binary e-shop flag and dropped usage value, giving clusters that are statistically tidy but useless for stocking.

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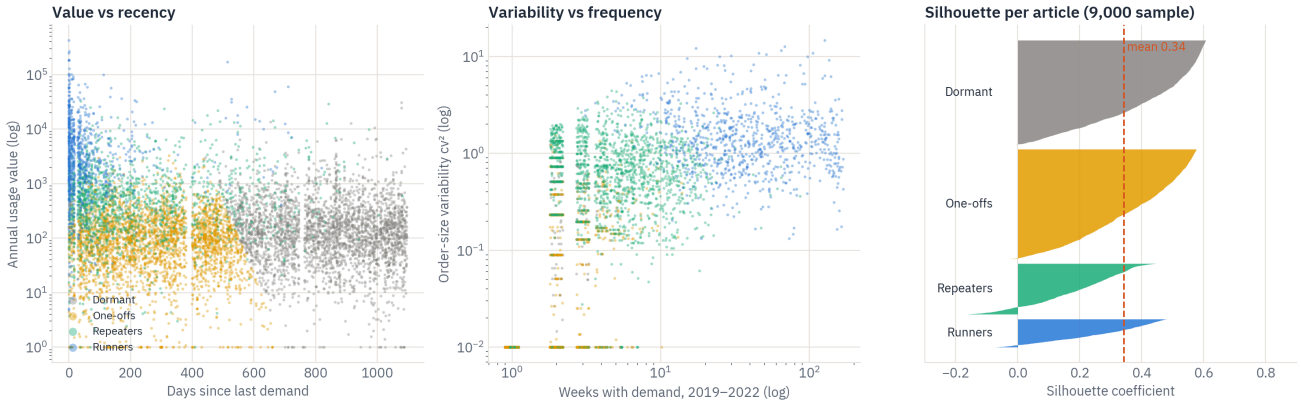
4 · The clusters, visualised

The four classes separate along two clear directions: how often and how much an article is used (left to right), and how long it has been idle (bottom to top).



Left: PCA map of all 20,890 active articles coloured by class, with class centres. Right: radar of each class's median percentile on six features

The first principal component (62% of variance) runs from rarely used, low-value articles to frequent, high-value, variable ones; the second (20%) is recency. Runners sit far right, Dormant articles top left, One-offs bottom left, and Repeaters in between. The radar shows that the classes also differ on features the model never saw: Dormant and One-off articles are the most expensive per unit, while Runners are cheap and often used in production.



Left: usage value vs days since last demand. Middle: order-size variability vs weeks with demand. Right: silhouette coefficient of each article, grouped by class

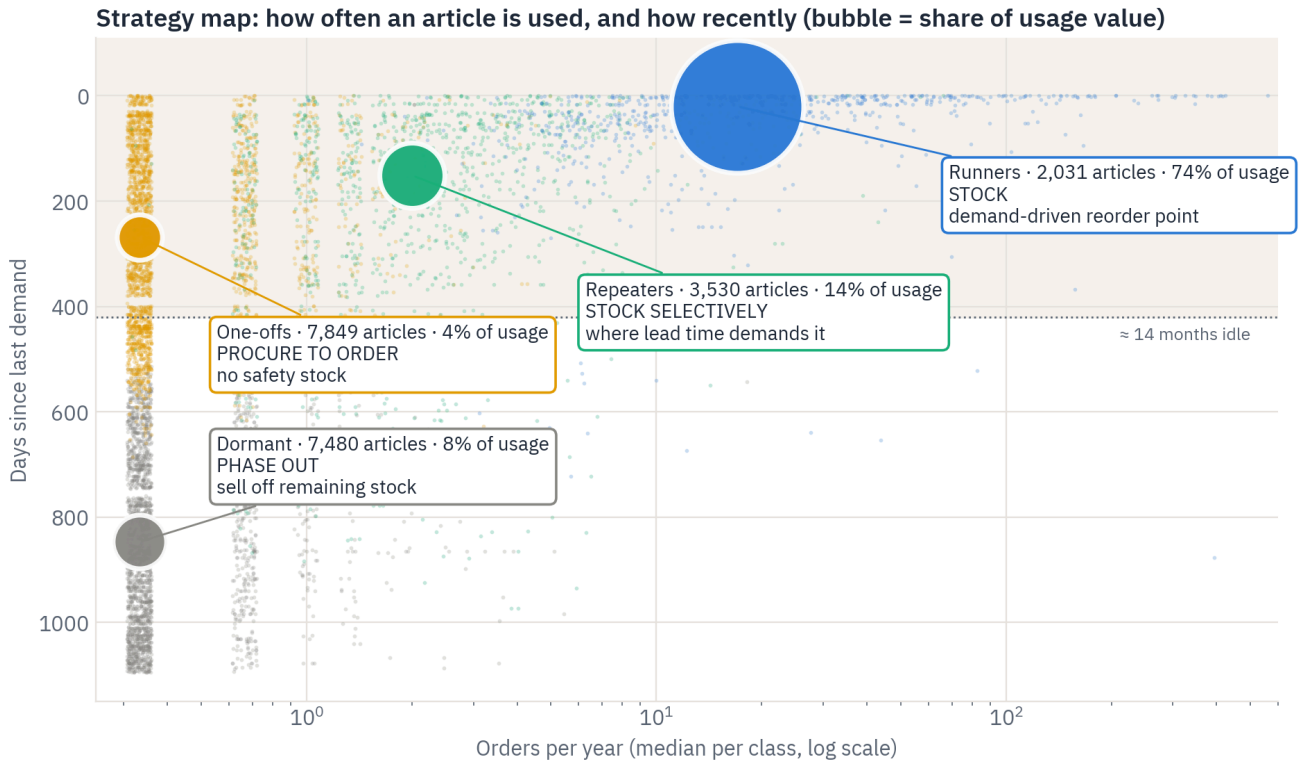
- **Value vs recency:** Runners cluster at recent, high-value demand; Dormant articles start where demand stopped about 14 months ago.
- **Variability vs frequency:** One-offs and Dormant articles have one or two orders (variability 0); Repeaters order a few times a year with variable sizes; Runners order weekly.

- **Silhouette:** mean 0.34. Dormant (0.41) and One-offs (0.37) are the cleanest classes; Repeaters (0.19) are the transition zone between occasional and regular demand, which is why their strategy is "selective".

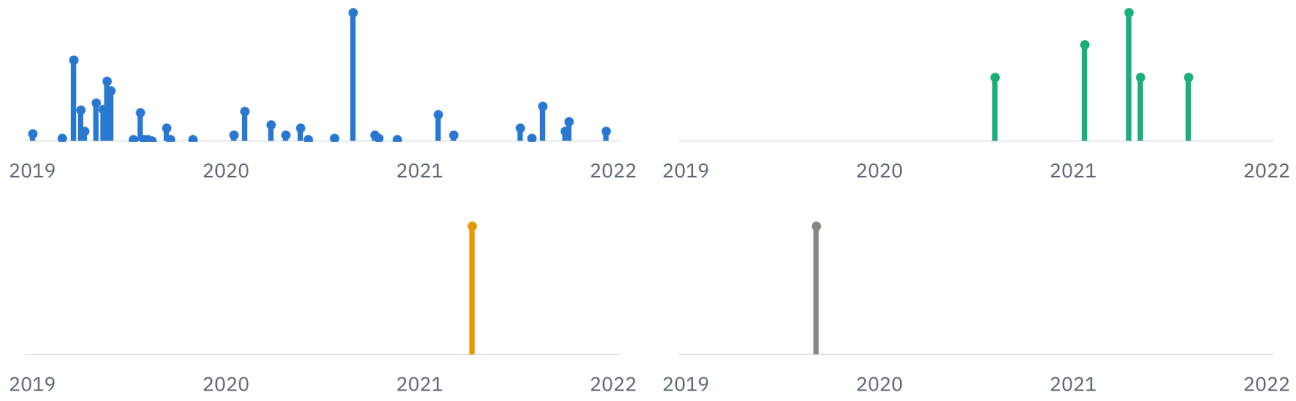
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5 · The four classes and their strategies

Only Runners, plus a selected part of the Repeaters, belong on stock; One-offs are bought to order and Dormant articles are phased out.



Strategy map: each class placed by median orders per year and days since last demand, bubble size = share of usage value, with the strategy for each class



Weekly demand 2019–2022 of one typical article per class: Runners (top left), Repeaters (top right), One-offs (bottom left), Dormant (bottom right)

Weekly demand of one typical article per class, 2019–2022: the Runner is ordered almost every few weeks, the Repeater a handful of times, the One-off and the Dormant article once. The strategy follows from that rhythm: a buffer pays off only where demand comes often enough to turn it over.

Class	Articles	Usage value	Stock value	Profile	Strategy	On stock?	Stocked today
Runners	2,031 (10%)	74%	79%	~17 orders a year, last used ~3 weeks ago, cheap, 37% production use	Stock with a demand-driven reorder point (section 6), service target by value	Yes	979
Repeaters	3,530 (17%)	14%	14%	~2 orders a year, lumpy sizes, last used ~5 months ago	Stock only where the supplier lead time is longer than customers will wait; otherwise to order	Selective	345
One-offs	7,849 (38%)	4%	5%	one order in three years, expensive per unit, last used ~9 months ago	Procure or produce to order; no safety stock	No	153
Dormant	7,480 (36%)	8%	2%	one order, no demand for 14+ months (median 2.3 years)	Phase out: stop replenishing, sell off or write down remaining stock	No	22

Stock value = average 2021 stock of the 20,890 active articles (7.45M). Usage value = annual usage, 2019–2022.

What changes for the client

- **De-list 427 stock articles:** 175 stocked One-offs and Dormant articles plus 252 stocked articles without any demand in three years.
- **Review 1,052 Runners without a reorder point.** Only 18% of them are mainly used in production, so most are candidates for proper stock management rather than ad-hoc buying.
- **Repeaters need a rule, not a default:** keep stock where the lead time exceeds the delivery time customers accept, with a lower service target (about 90%).
- **The class becomes master data:** recompute it every quarter, so articles that slow down move from Runners to Repeaters to Dormant automatically.

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6 · Reorder points and safety stock: why we simulated, and what it showed

A daily simulation of 2021 shows the demand-driven policy reaches 95% fill with 3.65M of average stock, against 4.92M actually held, and needs 34% less capital than today's reorder points at the same service.

Service levels: what A, B and C mean

Part 3 uses a second, separate segmentation. The Part 2 classes (Runners, Repeaters, One-offs, Dormant) decide **whether** an article belongs on stock, based on its demand behaviour. The ABC classes decide **how much safety buffer** a stocked article gets, based on its value.

ABC is the classic Pareto cut: rank the 1,751 inventory articles by annual usage value (12-month usage × purchase price), add up their shares, and cut at 80% and 95%.

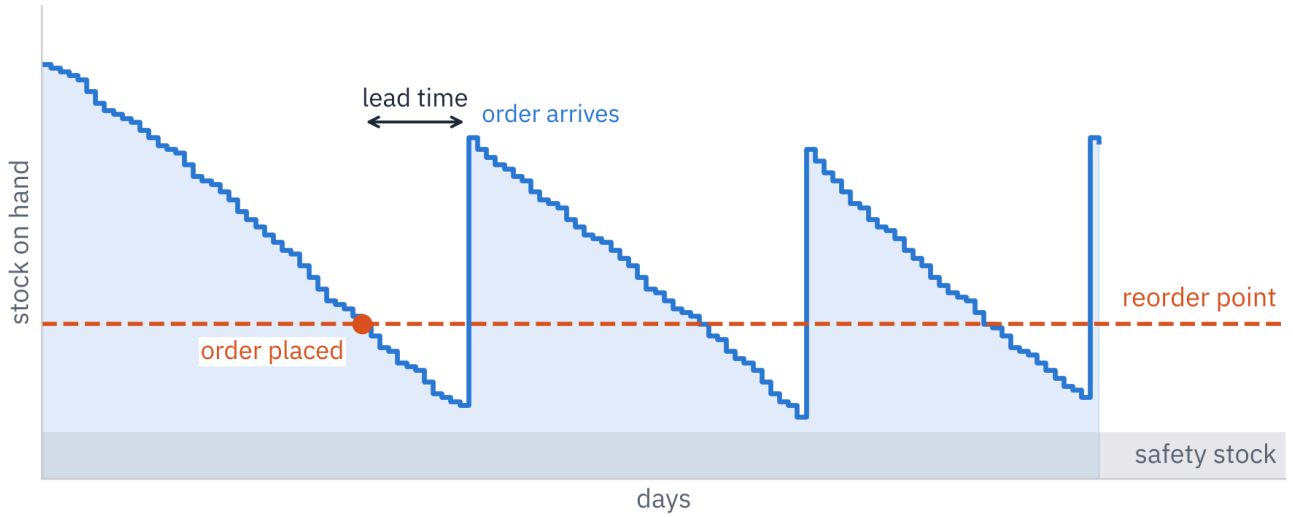
Class	Share of usage value	Articles	Fill-rate target
A	top 80%	167 (10%)	98%
B	next 15%	273 (16%)	95%
C	last 5%, plus articles without usage	1,311 (75%)	90%

A-articles hold about 65% of stock capital and a stockout there costs most, so they get the highest target and the pilot. C-articles are many but cheap, so a lower target saves capital at little risk. In the simulation, the safety parameter is calibrated per class so that each class just reaches its target. In one line: Part 2 decides *if* an article is stocked, ABC decides *how safely*.

Why simulation

- **A formula alone cannot be trusted here.** 75% of inventory articles have irregular demand, orders are large and the minimum order quantity binds, so textbook formulas give levels whose real effect is unknown.
- **History cannot be replayed directly.** Planners overrode the reorder points by hand, and recorded service is censored: usage is only booked when stock exists, so the actual 99.6% fill says little.
- **The client needs proof before changing 1,751 settings.** A simulation tests every policy on the same real demand, with the same lead times and order sizes, on a year the method never saw.

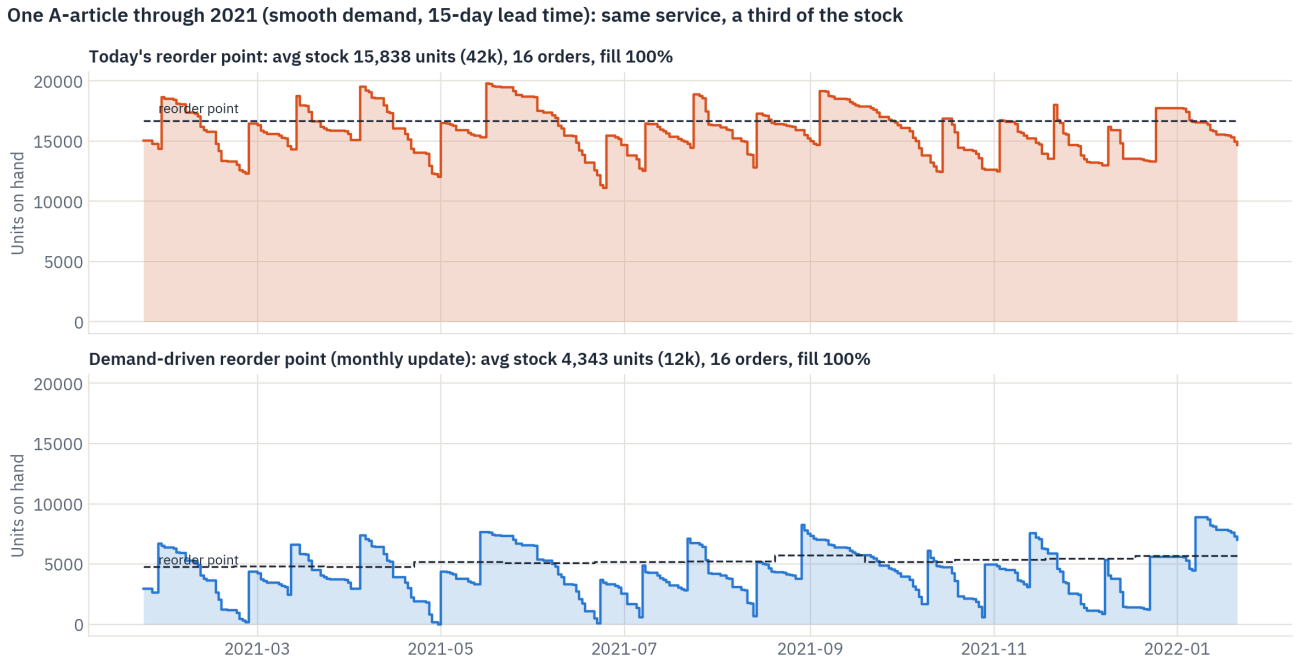
How the simulator works



Schematic of the simulated reorder policy: stock falls with daily demand; at the reorder point an order is placed and arrives after the lead time; safety stock is the buffer below

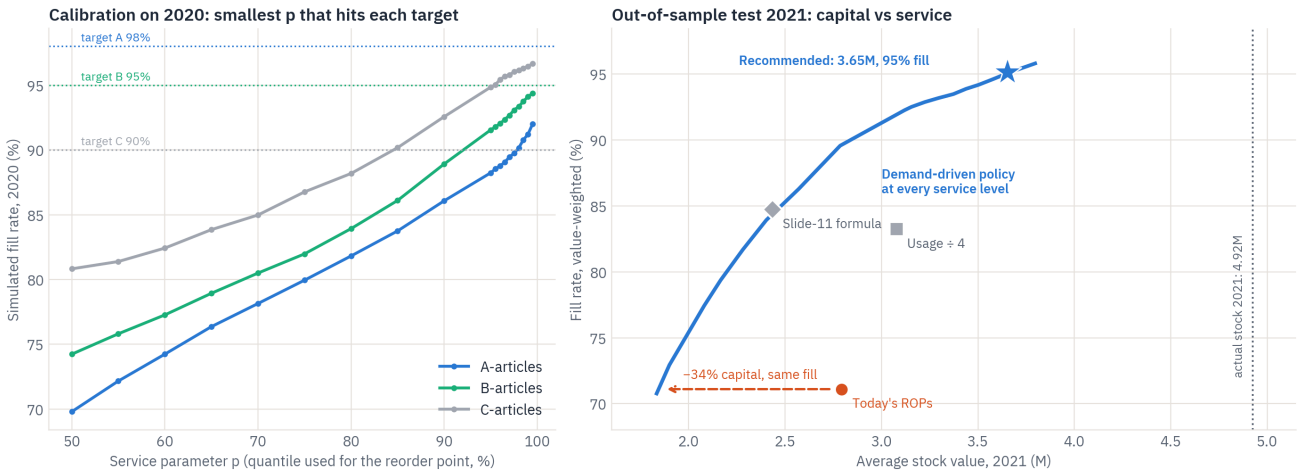
1. Every day, for all 1,751 inventory articles in parallel, base demand from 2021 is taken from stock; unmet demand is back-ordered.
2. When stock plus open orders falls to the reorder point, an order of today's typical size (at least the MOQ) is placed; it arrives after the article's lead time.

3. Rare large orders (daily demand above Q3 + 3·IQR, 5% of demand days, 29% of volume) are procured to order and not drawn from stock.
4. The demand-driven policy recalculates each reorder point monthly on the last 365 days: `μ·LT + z·σ·√(LT/7)` for steady demand, the p-quantile of past lead-time demand for irregular demand. Safety stock = reorder point − μ·LT.
5. Timeline without look-ahead: parameters from 2019–20, service level calibrated on 2020, results measured on 2021.



One A-article through 2021: daily stock on hand and reorder point under today's setting (top) and the demand-driven setting (bottom), same scale

The example makes the mechanism visible. Today's reorder point (16,661 units) is 3.2 times what this article needs, so it carries about 15,800 units on average. The demand-driven point follows demand month by month and carries about 4,300 units with the same 16 orders and 100% fill: 72% less capital on this one article.



Left: simulated fill rate on 2020 against the service parameter, per ABC class, with targets. Right: 2021 capital vs fill rate for the demand-driven policy and the current rules

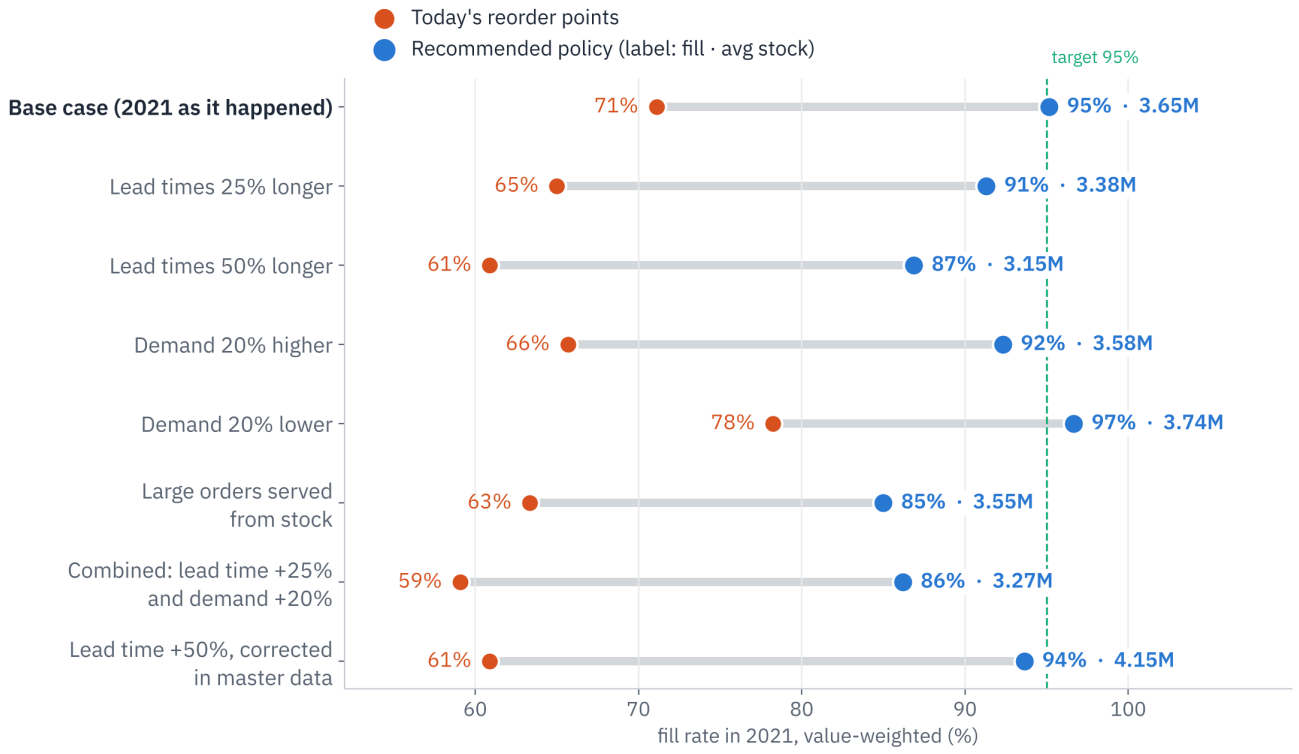
What it showed

Policy (2021, same simulator)	Avg stock	Fill rate	Fill A-articles
Actual 2021 (observed)	4.92M	–	–
Today's reorder points	2.79M	71%	68%
Rule: usage 12 months ÷ 4	3.08M	83%	83%
Case slide-11 formula	2.43M	85%	86%
Demand-driven, monthly update	**3.65M**	**95%**	**97%**

- **Calibration:** C-articles reach their 90% target at p = 85%. A and B reach the best attainable level (92% and 94% in 2020) at p = 99.5%; the gap comes from new, fast-growing articles that no history can anticipate.
- **Today's reorder points, followed strictly, would give only 71% fill**, which is why planners top up by hand and stock ended at 4.9M.

Stress test of the recommended strategy

A recommendation must survive bad lead times and demand shocks, so we re-ran the 2021 simulation under eight scenarios. Demand shocks are applied to 2021 only; the policy keeps recalculating its reorder points monthly.



Fill rate under eight scenarios, today's reorder points (orange) vs the recommended policy (blue, labelled with fill and average stock)

Scenario	Today's reorder points	Recommended: fill	Recommended: avg stock
Base case (2021 as it happened)	71%	**95%**	3.65M
Lead times 25% longer	65%	91%	3.38M
Lead times 50% longer	61%	87%	3.15M
Demand 20% higher	66%	92%	3.58M
Demand 20% lower	78%	97%	3.74M
Large orders served from stock	63%	85%	3.55M
Lead time +25% and demand +20%	59%	86%	3.27M
Lead time +50%, corrected in master data	61%	94%	4.15M

- The recommended policy **never falls below 85% fill** and stays **18 to 33 points ahead** of today's reorder points, which never reach 80%.
- **Lead times are the main risk.** If real lead times are 50% longer and nobody notices, fill drops to 87%; once corrected in the master data, the policy recovers to 94% at 4.15M, still below the 4.92M actually held.

What the simulation gave us

Exposed today's rules	followed strictly, today's reorder points deliver only 71% fill
Set the service levels	targets per ABC class reached at the least capital, chosen on 2020 data
Priced the impact	on 2021, a year the method never saw: −1.27M (−26%) at 95% fill
Tested the risks	eight stress scenarios, lead times are the data to fix first

How it helps the client: one tested rule replaces 1,751 manual settings; management can pick any point on the capital–service curve and know its cost in advance; and the stress tests show exactly which data (lead times) and processes (large orders) protect the result.

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7 · The strategy we recommend

Stock the right articles, at levels set by demand: clean the range, decide stocking by class, and let a monthly demand-driven rule set every reorder point and safety stock.

1. **Clean the range (months 1–2).** De-list 427 stock articles (One-offs, Dormant, no demand in three years); stop replenishing them and sell off or write down 0.94M of stock idle for over a year.
2. **Stock by class (months 1–3).** Runners on stock; Repeaters only where the supplier lead time is longer than customers will wait; One-offs procured to order; Dormant phased out. Review the 1,052 Runners without a reorder point. Recompute the classes every quarter.
3. **Set levels from demand (months 2–6).** Replace manual reorder points with the demand-driven rule, recalculated monthly, with service targets A 98% / B 95% / C 90%.

Start with the A-articles (about 10% of articles, 65% of capital) as a pilot, compare with the simulation, then roll out. Every article's recommended level is in `Part3_ROP_SafetyStock_Recommendations.xlsx`.

4. **Protect the result (ongoing).** Record order and receipt dates to measure real lead times (a 50% error costs 8 points of fill); plan rare large orders through customer and production orders, not stock; set order size and reorder point together, since order size is the strongest driver of the right level.

Roadmap

Step	M1	M2	M3	M4	M5	M6
1 · Clean the range	■	■				
2 · Stock by class	■	■	■			
3 · Levels from demand (A-article pilot first)		■	■	■	■	■
4 · Protect the result	---	---	---	---	---	---

Expected impact

Measure	Today	With the strategy
Average stock, inventory articles	4.92M (2021)	3.65M (−1.27M, −26%)
Fill rate, value-weighted	71% if today's reorder points were followed	95% (A-articles 97%)
Holding cost (20% p.a., assumed)	–	about 0.25M a year saved (0.19–0.32M at 15–25%)
Idle stock to clear	0.94M idle > 1 year	sold off or written down once
Reorder points to maintain by hand	1,751	0: one rule, recalculated monthly

8 · Limitations and next steps

The results are robust to the main data gaps, but three assumptions should be validated with the client before roll-out.

- **Master data is a 2022 snapshot.** History is valued at today's purchase price and compared with today's settings.
- **No order dates,** so lead-time variability cannot be measured; the stress test (+50% lead time → 87% fill) bounds the risk.
- **Holding cost rate (20%) is an assumption.** Capital figures do not depend on it; only the yearly saving does.
- **Censored service history:** the recorded 99.6% fill reflects usage booked only when stock exists, so service is compared inside the simulation, not against history.
- **Values are anonymised:** quantities carry a constant factor (×1.1107) and money is in dataset currency units; ratios and percentages are unaffected.

Next steps: agree service targets per class with the client, validate lead times for the A-articles, run the A-article pilot for one quarter, and quantify the capital released by phasing out One-offs and Dormant articles.

Reproducibility: all analysis runs from the raw file in the project repository github.com/divyashori-datascience/scm-inventory-optimisation: `shared/prep.py` for cleaning, then `part1_data_analysis`, `part2_article_classification` and `part3_safety_stock_rop`, each with its own README and notebooks.